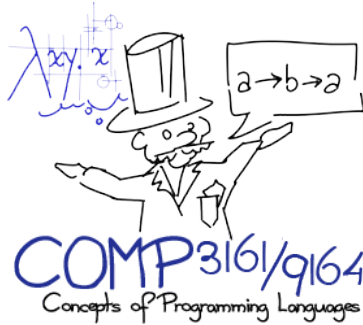




Natural Deduction and Rule Induction

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Formalisation

To talk about languages in a mathematically precise way, we need to **formalise** them.

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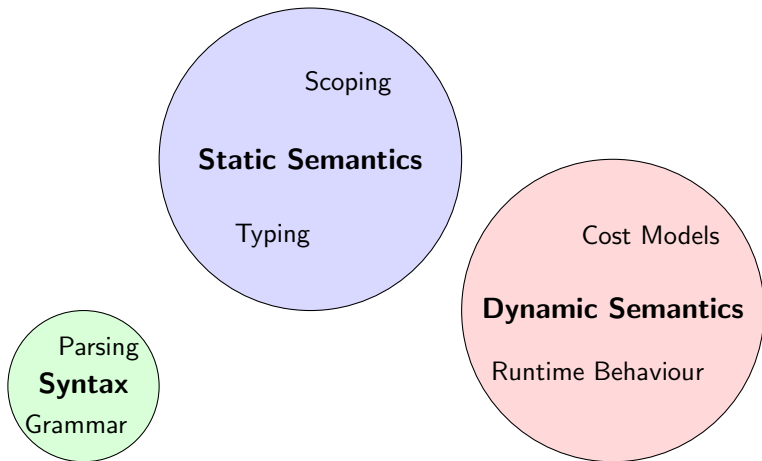
Formalisation

Formalisation is the process of giving a language a formal, **mathematical description**.

Typically, we describe the language in **another language**, called the *meta-language*. For implementations, it may be a programming language such as **Haskell**. For formalisations it is usually a minimal logic called a *meta-logic*.

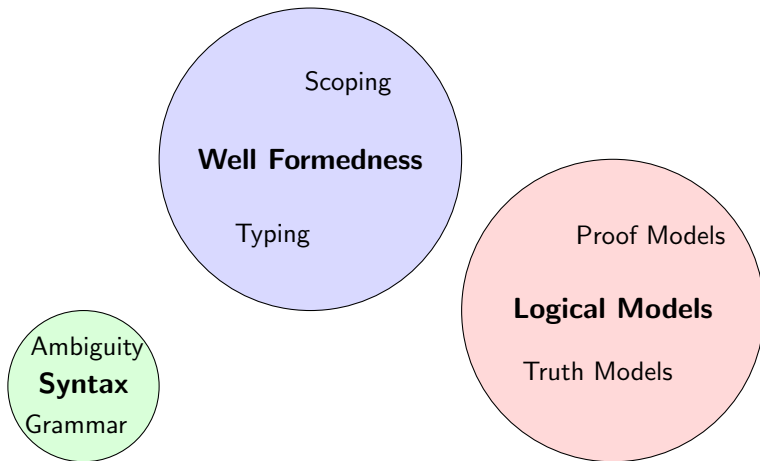
Learning from History

What sort of meta logic should we use? There are a number of things to formalise:



Learning from History

Logicians in the early 20th century had much the same desire to formalise *logics*.



Learning from History

In this course, we will use a meta-logic based on *Natural Deduction* and inductive inference rules, originally invented for formalising logics by Gerhard Gentzen in the mid 1930s.

Der Kalkül des natürlichen Schließens.

$$\begin{array}{ccc}
 \frac{\mathcal{A} \quad \mathcal{B}}{\mathcal{A} \& \mathcal{B}} & \frac{\mathcal{A} \& \mathcal{B}}{\mathcal{A}} & \frac{\mathcal{A} \& \mathcal{B}}{\mathcal{B}}
 \end{array}$$

Judgements

A *judgement* is a statement asserting a certain property for an object.

Example (Informal Judgements)

- $3 + 4 \times 5$ is a valid arithmetic expression.
- The string *madam* is a palindrome.
- The string *snooze* is a palindrome
 $\implies$ Judgements do not have to hold.

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- $\implies$ Judgements do not have to hold.

Unary Judgements

Formally, we denote the judgement that a property **A** holds for an object *s* by writing *s A*.

Typically, *s* is a *string* when describing syntax, and *s* is a *term* when describing semantics.

Proving Judgements

We define how a judgement may be **proven** by providing a set of *inference rules*.

Inference Rules

An inference rule is written as:

$$\frac{J_1 \quad J_2 \quad \dots \quad J_n}{J}$$

This states that in order to prove judgement J (the *conclusion*), it suffices to prove all judgements J_1 through to J_n (the *premises*).

Rules with no premises are called *axioms*. Their conclusions **always hold**.

Examples

Example (Natural Numbers)

$n \text{ Nat}$

$$\frac{}{0 \text{ Nat}} N_1$$

0 is a natural number

$$\frac{n \text{ Nat}}{(S \ n) \text{ Nat}} N_2$$

if n is a natural number,
then the successor of n
is a natural number.

What terms are in the set $\{n \mid n \text{ Nat}\}$?

Examples

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if n is a natural number,
then the successor of n
is a natural number.

What terms are in the set $\{n \mid n \text{ Nat}\}$?

$$\{0, (S \ 0), (S \ (S \ 0)), (S \ (S \ (S \ 0))), \dots\}$$

Examples

Example (Even and Odd Numbers)

$$\begin{array}{ccc}
 \boxed{n \text{ Even}} & \boxed{n \text{ Odd}} & \\
 \frac{}{0 \text{ Even}} E_1 & \frac{n \text{ Even}}{(S (S n)) \text{ Even}} E_2 & \frac{n \text{ Even}}{(S n) \text{ Odd}} O_1
 \end{array}$$

The Proof Video Game

To show that a judgement $s \mathbf{A}$ holds:

- ① Find a rule whose conclusion matches $s \mathbf{A}$.
- ② The preconditions of the applied rules become new **proof obligations**.
- ③ Rinse and repeat until all obligations are proven up to axioms.

Examples

Example (Even and Odd Numbers)

$$\begin{array}{ccc}
 \boxed{n \text{ Even}} & \boxed{n \text{ Odd}} & \\
 \hline
 0 \text{ Even} & \frac{n \text{ Even}}{(S (S n)) \text{ Even}} & \frac{n \text{ Even}}{(S n) \text{ Odd}} \\
 E_1 & E_2 & O_1
 \end{array}$$

$$\overline{(S (S (S (S (S 0)))) \text{ Odd}}$$

Examples

Example (Even and Odd Numbers)

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 E_1 & E_2 & O_1
 \end{array}$$

$$\frac{(S (S (S (S 0)))) \text{ Even}}{(S (S (S (S (S 0)))) \text{ Odd}} O_1$$

Examples

Example (Even and Odd Numbers)

$$\begin{array}{ccc}
 \boxed{n \text{ Even}} & \boxed{n \text{ Odd}} & \\
 \hline
 0 \text{ Even} & \frac{n \text{ Even}}{(S (S n)) \text{ Even}} E_2 & \frac{n \text{ Even}}{(S n) \text{ Odd}} O_1
 \end{array}$$

$\frac{}{0 \text{ Even}} E_1$

$$\frac{\frac{\frac{(S (S 0)) \text{ Even}}{(S (S (S (S 0)))) \text{ Even}} E_2}{(S (S (S (S (S 0)))) \text{ Odd}} O_1}$$

Examples

Example (Even and Odd Numbers)

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 \boxed{n \text{ Even}} & \boxed{n \text{ Odd}} & \\
 \hline
 0 \text{ Even} & \frac{n \text{ Even}}{(S (S n)) \text{ Even}} E_2 & \frac{n \text{ Even}}{(S n) \text{ Odd}} O_1
 \end{array}$$

$$\begin{array}{c}
 \hline
 0 \text{ Even} \\
 \hline
 (S (S 0)) \text{ Even} E_2 \\
 \hline
 (S (S (S (S 0)))) \text{ Even} E_2 \\
 \hline
 (S (S (S (S (S 0)))) \text{ Odd} O_1
 \end{array}$$

Examples

Example (Even and Odd Numbers)

$$\begin{array}{ccc}
 \boxed{n \text{ Even}} & \boxed{n \text{ Odd}} & \\
 \frac{}{0 \text{ Even}} E_1 & \frac{n \text{ Even}}{(S (S n)) \text{ Even}} E_2 & \frac{n \text{ Even}}{(S n) \text{ Odd}} O_1
 \end{array}$$

$$\frac{\frac{\frac{\frac{\frac{}{0 \text{ Even}} E_1}{(S (S 0)) \text{ Even}} E_2}{(S (S (S (S 0)))) \text{ Even}} E_2}{(S (S (S (S (S 0)))) \text{ Odd}} O_1$$

Defining Languages

Example (Bracket Matching Language)

$$\mathbf{M} ::= \varepsilon \mid \mathbf{MM} \mid (\mathbf{M})$$

Examples of strings: ε , $()$, $(())$, $()()$, $((()))$, ...

Three rules:

Axiom The empty string is in **M**

Juxtaposition Any two strings in **M** can be concatenated to give a new string in **M**

Nesting Any string in **M** can be surrounded by parentheses, giving a new string in **M**

With Rules

The Language M

$$\begin{array}{c}
 \boxed{s \ M} \\
 \\
 \frac{}{\varepsilon \ M} M_E \qquad \frac{s \ M}{(s) \ M} M_N \qquad \frac{s_1 \ M \quad s_2 \ M}{s_1 s_2 \ M} M_J
 \end{array}$$

$((())) \ M$



With Rules

The Language M

$$\begin{array}{c}
 \boxed{s \ M} \\
 \\
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 \end{array}$$

$$\frac{\frac{}{() \ M} \quad \frac{}{(()) \ M}}{() (()) \ M} M_J$$

With Rules

The Language M

s M

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$$\frac{s_1 \mathbf{M} \quad s_2 \mathbf{M}}{s_1 s_2 \mathbf{M}} M_J$$

$$\frac{\frac{\frac{}{\varepsilon \mathbf{M}} M_E}{() \mathbf{M}} M_N \quad \frac{}{(()) \mathbf{M}}}{() (()) \mathbf{M}} M_J$$

With Rules

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$$\frac{\frac{\frac{}{\varepsilon \mathbf{M}} M_E}{() \mathbf{M}} M_N \quad \frac{\frac{\frac{}{\varepsilon \mathbf{M}} M_E}{() \mathbf{M}} M_N}{(()) \mathbf{M}} M_N}{() (()) \mathbf{M}} M_J$$

Getting Stuck

If we had started with rule M_N instead, we would have gotten stuck:

$$\frac{\frac{???}{) (() \mathbf{M}}}{((()) \mathbf{M}} M_N$$

Takeaway

Getting stuck does **not** mean what you're trying to prove is false!

Derivability

Consider the following rule:

$$\frac{s \text{ M}}{((s)) \text{ M}}$$

Does adding this rule change **M**? (i.e. is it not *admissible* to **M**)?

Derivability

Consider the following rule:

$$\frac{s \ M}{((s)) \ M}$$

Does adding this rule change **M**? (i.e. is it not *admissible* to **M**)?
No, because we could always use rule M_N twice instead. Rules that are compositions of existing rules are called *derivable*:

$$\frac{\frac{s \ M}{(s) \ M} M_N}{((s)) \ M} M_N$$

We can prove **rules** as well as **judgements**, by deriving the **conclusion** of the rule while taking the **premises** as local axioms.



Derivability

Is this rule derivable?

$$\frac{s \quad M}{(s) s \quad M}$$



Derivability

Is this rule derivable?

$$\frac{s \ M}{(s) s \ M}$$

We can derive it like so:

$$\frac{\frac{\overline{s \ M}}{(s) \ M} M_N \quad \frac{\overline{s \ M}}{s \ M} M_J}{(s) s \ M}$$



Derivability

Is this rule derivable?

$$\frac{(s) \text{ M}}{s \text{ M}} Q$$

Derivability

Is this rule admissible? If so, is it derivable?

$$\frac{()s \text{ M}}{s \text{ M}}$$



Derivability

Is this rule admissible? If so, is it derivable?

$$\frac{()s \text{ M}}{s \text{ M}}$$

- It is **admissible**, as it doesn't let us prove any new judgements about **M**.
- It is **not derivable**, as it is not made up of the composition of existing rules.
- We will see how to prove these sorts of rules are admissible later on.

Hypothetical Derivations

We can write a rule in a horizontal format as well:

$$\frac{A}{B} \text{ is the same as } A \vdash B$$

This allows us to neatly make **rules** premises of other rules, called *hypothetical derivations*:

Example

$$\frac{A \vdash B}{C}$$

Read as: *If assuming A we can derive B , then we can derive C .*

Specifying Logic

With hypotheticals we can specify logic, which was the original purpose of natural deduction. Let A **True** be the judgement that the proposition A is true.

Example (And and Implies)

$$\frac{A \text{ True} \quad B \text{ True}}{A \wedge B \text{ True}} \wedge_I \quad \frac{A \wedge B \text{ True}}{A \text{ True}} \wedge_{E1} \quad \frac{A \wedge B \text{ True}}{B \text{ True}} \wedge_{E2}$$

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 \\
 \frac{A \text{ True} \vdash B \text{ True}}{A \Rightarrow B \text{ True}} \Rightarrow_I \quad \frac{A \Rightarrow B \text{ True} \quad A \text{ True}}{B \text{ True}} \Rightarrow_E
 \end{array}$$



Specifying Logic, Continued

Example (Or, True, False and Not)

$$\begin{array}{c}
 \frac{A \text{ True}}{A \vee B \text{ True}} \vee_{I1} \quad \frac{B \text{ True}}{A \vee B \text{ True}} \vee_{I2} \\
 \frac{A \text{ True} \vdash C \text{ True} \quad B \text{ True} \vdash C \text{ True} \quad A \vee B \text{ True}}{C \text{ True}} \vee_E
 \end{array}$$

Specifying Logic, Continued

Example (Or, True, False and Not)

$$\begin{array}{c}
 \frac{A \text{ True}}{A \vee B \text{ True}} \vee_{I1} \quad \frac{B \text{ True}}{A \vee B \text{ True}} \vee_{I2} \\
 \\
 \frac{A \text{ True} \vdash C \text{ True} \quad B \text{ True} \vdash C \text{ True} \quad A \vee B \text{ True}}{C \text{ True}} \vee_E \\
 \\
 \frac{}{\top \text{ True}} \top_I \quad \frac{\perp \text{ True}}{A \text{ True}} \perp_E \\
 \\
 \frac{A \text{ True} \vdash \perp \text{ True}}{\neg A \text{ True}} \neg_I \quad \frac{\neg A \text{ True} \quad A \text{ True}}{B \text{ True}} \neg_E
 \end{array}$$

Minimal Definitions

$$\begin{array}{c}
 \boxed{s \text{ M}} \\
 \\
 \frac{}{\varepsilon \text{ M}} M_E \qquad \frac{s \text{ M}}{(s) \text{ M}} M_N \qquad \frac{s_1 \text{ M} \quad s_2 \text{ M}}{s_1 s_2 \text{ M}} M_J
 \end{array}$$

The above rules are the **smallest set of rules** to define every string in **M**.

Therefore

If we know that a string satisfies **s M**, it must have been through a (finite) derivation using these rules.

This is called an **inductive definition** of **M**.

Rule Induction

Suppose we want to show that a property $P(s)$ of strings s holds for any string s **M**. We will use *rule induction*.

If we show that

$$\frac{}{\varepsilon \text{ **M**}} M_E$$

$P(\varepsilon)$ holds, and

$$\frac{s \text{ **M**}}{(s) \text{ **M**}} M_N$$

$P(s)$ implies $P((s))$, and

$$\frac{s_1 \text{ **M**} \quad s_2 \text{ **M**}}{s_1 s_2 \text{ **M**}} M_J$$

$P(s_1)$ and $P(s_2)$ implies $P(s_1 s_2)$

Then we have shown $P(s)$ for all s **M**.

These assumptions are called *inductive hypotheses*.

Rule Induction

Example (Counting Parens)

Let $op(s)$ denote the number of opening parentheses in s , and $cl(s)$ denote the number of closing parentheses. We shall prove that

$$s \mathbf{M} \implies op(s) = cl(s)$$

by doing rule induction on $s \mathbf{M}$.



Rule Induction

Example (Counting Parens)

$$\frac{}{\varepsilon \mathbf{M}} M_E$$

Base Case: $op(\varepsilon) = 0 = cl(\varepsilon)$

Rule Induction

Example (Counting Pairs)

$$\frac{}{\varepsilon \text{ M}} M_E$$

$$\frac{s \text{ M}}{(s) \text{ M}} M_N$$

Base Case: $op(\varepsilon) = 0 = cl(\varepsilon)$

Inductive Case: Assuming I.H:

$$op(s) = cl(s)$$

$$op((s)) = op(s) + 1 = cl(s) + 1 = cl((s))$$

Rule Induction

Example (Counting Prens)

$$\frac{}{\varepsilon \text{ M}} M_E$$

$$\frac{s \text{ M}}{(s) \text{ M}} M_N$$

$$\frac{s_1 \text{ M} \quad s_2 \text{ M}}{s_1 s_2 \text{ M}} M_J$$

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Inductive Case: Assuming I.H:

$$op(s) = cl(s)$$

$$op((s)) = op(s) + 1 = cl(s) + 1 = cl((s))$$

Inductive Case: Assuming I.Hs:

$$op(s_1) = cl(s_1) \text{ and } op(s_2) = cl(s_2)$$

$$op(s_1 s_2) = op(s_1) + op(s_2) = cl(s_1 s_2)$$

Rule Induction in General

Rule Induction Method

Given a set of rules R , we may prove a property P **inductively** for all judgements that can be inferred with R by showing, for each rule of the form

$$\frac{J_1 \quad J_2 \quad \dots \quad J_n}{J}$$

that if P holds for each of $J_1 \dots J_n$, then P holds for J .

Therefore, axioms are the **base cases** of the induction, all other rules form **inductive cases**, and the premises of each rule give rise to **inductive hypotheses**.

Structural Induction

Conventional *structural induction* such as that on natural numbers, which we have encountered before, is a **special case** of rule induction.

Natural Number Induction

To show a property $P(n)$ for all $n \in \mathbb{N}$, it suffices to:

$\frac{}{0 \text{ Nat}}$ Show that $P(0)$ holds, and

$\frac{n \text{ Nat}}{(S \ n) \text{ Nat}}$ Assuming $P(n)$, show $P(n + 1)$.

Another Example

Recall our definition of even numbers:

$$\begin{array}{c}
 \boxed{n \text{ Even}} \\
 \frac{}{0 \text{ Even}} E_1 \qquad \frac{n \text{ Even}}{(S (S n)) \text{ Even}} E_2
 \end{array}$$

We could define odd numbers differently:

$$\begin{array}{c}
 \boxed{n \text{ Odd}'} \\
 \frac{}{(S 0) \text{ Odd}'} O'_1 \qquad \frac{n \text{ Odd}'}{(S (S n)) \text{ Odd}'} O'_2
 \end{array}$$

Let's prove the original **Odd** rule, but for **Odd'** (to “whiteboard”):

$$\frac{n \text{ Even}}{(S n) \text{ Odd}'}$$

Arithmetic

Example (Arithmetic Expression)

Arith ::= i | **Arith** $\times$ **Arith** | **Arith** $+$ **Arith** | (**Arith**) ($i \in \mathbb{Z}$)

Arithmetic

Example (Arithmetic Expression)

Arith ::= i | **Arith** $\times$ **Arith** | **Arith** $+$ **Arith** | (**Arith**) ($i \in \mathbb{Z}$)

$$\frac{i \in \mathbb{Z}}{i \text{ Arith}} L \quad \frac{a \text{ Arith} \quad b \text{ Arith}}{a \times b \text{ Arith}} P \quad \frac{a \text{ Arith} \quad b \text{ Arith}}{a + b \text{ Arith}} S \quad \frac{a \text{ Arith}}{(a) \text{ Arith}}$$

We can infer $1 + 2 \times 3$ **Arith** in two different ways.

Ambiguity

Arith is *ambiguous*, which means that there are multiple ways to derive the same judgement.

For syntax, this is a **big problem**, as different interpretations of syntax can lead to semantic inconsistency:

$$\begin{array}{c}
 \frac{1 \in \mathbb{Z}}{1 \text{ Arith}} \quad \frac{\frac{2 \in \mathbb{Z}}{2 \text{ Arith}} \quad \frac{3 \in \mathbb{Z}}{3 \text{ Arith}}}{2 \times 3 \text{ Arith}} \\
 \hline
 1 + 2 \times 3 \text{ Arith}
 \end{array}
 \qquad
 \begin{array}{c}
 \frac{1 \in \mathbb{Z}}{1 \text{ Arith}} \quad \frac{2 \in \mathbb{Z}}{2 \text{ Arith}} \quad \frac{3 \in \mathbb{Z}}{3 \text{ Arith}} \\
 \frac{1 + 2 \text{ Arith}}{1 + 2 \times 3 \text{ Arith}}
 \end{array}$$

Second Attempt

We want to specify **Arith** in such a way that enforces **order of operations**.

Here we will use **multiple judgements**:

Example (Arithmetic Expression)

Atom ::= $i \mid (\text{SExp}) \quad (i \in \mathbb{Z})$

PExp ::= **Atom** $\mid$ **PExp** $\times$ **PExp**

SExp ::= **PExp** $\mid$ **SExp** $+$ **SExp**

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PExp ::= **Atom** $\mid$ **PExp** $\times$ **PExp**

SExp ::= **PExp** $\mid$ **SExp** $+$ **SExp**

$$\begin{array}{c}
 \frac{i \in \mathbb{Z}}{i \text{ Atom}} \quad \frac{a \text{ SExp}}{(a) \text{ Atom}} \quad \frac{e \text{ Atom}}{e \text{ PExp}} \quad \frac{e \text{ PExp}}{e \text{ SExp}} \\
 \frac{a \text{ PExp} \quad b \text{ PExp}}{a \times b \text{ PExp}} \quad \frac{a \text{ SExp} \quad b \text{ SExp}}{a + b \text{ SExp}}
 \end{array}$$

Consider: Is there still any ambiguity here?

More ambiguity

$$\begin{array}{c}
 \frac{1 \in \mathbb{Z}}{1 \text{ Atom}} \quad \frac{\frac{2 \in \mathbb{Z}}{2 \text{ Atom}} \quad \frac{3 \in \mathbb{Z}}{3 \text{ Atom}}}{2 \text{ PExp} \quad 3 \text{ PExp}} \\
 \frac{1 \text{ PExp} \quad 2 \times 3 \text{ PExp}}{1 \times 2 \times 3 \text{ PExp}}
 \end{array}
 \qquad
 \begin{array}{c}
 \frac{1 \in \mathbb{Z}}{1 \text{ Atom}} \quad \frac{2 \in \mathbb{Z}}{2 \text{ Atom}} \quad \frac{3 \in \mathbb{Z}}{3 \text{ Atom}} \\
 \frac{1 \text{ PExp} \quad 2 \text{ PExp}}{1 \times 2 \text{ PExp}} \quad \frac{3 \text{ PExp}}{3 \text{ PExp}} \\
 \frac{1 \times 2 \text{ PExp} \quad 3 \text{ PExp}}{1 \times 2 \times 3 \text{ PExp}}
 \end{array}$$

This ambiguity seems harmless, but it would not be harmless for some other operations. Which ones?

More ambiguity

$$\begin{array}{c}
 \frac{1 \in \mathbb{Z}}{1 \text{ Atom}} \quad \frac{\frac{2 \in \mathbb{Z}}{2 \text{ Atom}} \quad \frac{3 \in \mathbb{Z}}{3 \text{ Atom}}}{2 \text{ PExp} \quad 3 \text{ PExp}} \\
 \frac{1 \text{ PExp} \quad 2 \times 3 \text{ PExp}}{1 \times 2 \times 3 \text{ PExp}}
 \end{array}
 \qquad
 \begin{array}{c}
 \frac{1 \in \mathbb{Z}}{1 \text{ Atom}} \quad \frac{2 \in \mathbb{Z}}{2 \text{ Atom}} \quad 3 \in \mathbb{Z} \\
 \frac{1 \text{ PExp} \quad 2 \text{ PExp}}{1 \times 2 \text{ PExp}} \quad \frac{3 \text{ Atom}}{3 \text{ PExp}} \\
 \frac{1 \times 2 \text{ PExp} \quad 3 \text{ PExp}}{1 \times 2 \times 3 \text{ PExp}}
 \end{array}$$

This ambiguity seems harmless, but it would not be harmless for some other operations. Which ones? Operators that are not *associative*.

We have to specify the *associativity* of operators. How?

Associativities

Operators have various *associativity* constraints:

Associative

All associativities are equal.

Left-Associative

$$A \odot B \odot C = (A \odot B) \odot C$$

Right-Associative

$$A \odot B \odot C = A \odot (B \odot C)$$

Try to think of some examples!



Enforcing associativity

We force the grammar to accept a smaller set of expressions on **one** side of the operator only. Show how this works on the “whiteboard”.

Example (Arithmetic Expression)

$$\begin{aligned}
 \text{Atom} &::= i \mid (\text{SExp}) \quad (i \in \mathbb{Z}) \\
 \text{PExp} &::= \text{Atom} \mid \text{Atom} \times \text{PExp} \\
 \text{SExp} &::= \text{PExp} \mid \text{PExp} + \text{SExp}
 \end{aligned}$$

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We force the grammar to accept a smaller set of expressions on **one** side of the operator only. Show how this works on the “whiteboard”.

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Atom ::= $i \mid (\text{SExp}) \quad (i \in \mathbb{Z})$

PExp ::= **Atom** $\mid$ **Atom** $\times$ **PExp**

SExp ::= **PExp** $\mid$ **PExp** $+$ **SExp**

$$\begin{array}{c}
 \frac{i \in \mathbb{Z}}{i \text{ Atom}} \quad \frac{a \text{ SExp}}{(a) \text{ Atom}} \quad \frac{e \text{ Atom}}{e \text{ PExp}} \quad \frac{e \text{ PExp}}{e \text{ SExp}} \\
 \\
 \frac{a \text{ Atom} \quad b \text{ PExp}}{a \times b \text{ PExp}} \quad \frac{a \text{ PExp} \quad b \text{ SExp}}{a + b \text{ SExp}}
 \end{array}$$

Here we made multiplication and addition **right** associative. How would we do **left**?

Bring Back Parentheses

The Parenthetical Language

$$\begin{array}{c}
 \boxed{s \ M} \\
 \frac{}{\varepsilon \ M} M_E \qquad \frac{s \ M}{(s) \ M} M_N \qquad \frac{s_1 \ M \quad s_2 \ M}{s_1 s_2 \ M} M_J
 \end{array}$$

Is this language ambiguous? to “whiteboard”

Ambiguity in Parentheses

Not only is it ambiguous, it is **infinitely** so. Strings like $()()()$ could be split at two different locations by rule M_J , but if we use ε , then even the string $()$ is ambiguous:

$$\begin{array}{c}
 \frac{\frac{\varepsilon \mathbf{M}}{\quad} M_E}{() \mathbf{M}} M_N \qquad \frac{\frac{\varepsilon \mathbf{M}}{\quad} M_E \quad \frac{\frac{\varepsilon \mathbf{M}}{\quad} M_E}{() \mathbf{M}} M_N}{() \mathbf{M}} M_J
 \end{array}$$

$$\frac{\frac{\varepsilon \mathbf{M}}{\quad} M_E \quad \frac{\frac{\varepsilon \mathbf{M}}{\quad} M_E \quad \frac{\frac{\varepsilon \mathbf{M}}{\quad} M_E}{() \mathbf{M}} M_N}{() \mathbf{M}} M_J}{() \mathbf{M}} M_J$$

We will eliminate the ambiguity by once again splitting **M** into two judgements, **N** and **L**.

The crucial observation is that terms in **M** are a **list** (**L**) of terms nested within parentheses (**N**).

Example (Unambiguous Parentheses)

$$\begin{array}{ccc}
 \boxed{s \ L} & & \boxed{s \ N} \\
 \\
 \frac{}{\varepsilon \ L} L_E & \quad \quad & \frac{s \ L}{(s) \ N} N_N \quad \quad \quad \frac{s_1 \ N \quad s_2 \ L}{s_1 s_2 \ L} L_J
 \end{array}$$



Proving Equivalence

Now we shall prove $\mathbf{M} = \mathbf{L}$. There are two cases, each dispatched with rule induction:

$$\frac{s \mathbf{M}}{s \mathbf{L}} \quad \frac{s \mathbf{L}}{s \mathbf{M}}$$

The first case requires proving a *lemma*. The second requires *simultaneous induction*.

These proofs will be carried out on the “board”.